Raman Spectroscopy

Professor C V Raman received the Nobel Prize in Physics in 1930 for his work done from The Indian Association for the Cultivation of Science (IACS) on the scattering of light and for the discovery of the effect named after him.

SIR CHANDRASEKHARA V. RAMAN

The molecular scattering of light

Nobel Lecture, December 11, 1930

The chief difficulty which had hitherto oppressed us in the study of the new phenomenon was its extreme feebleness in general. This was overcome by using a 7-inch refracting telescope in combination with a short-focus lens to condense sunlight into a pencil of very great intensity. With these arrangements and using complementary light-filters in the path of the incident and scattered beams, as was done by Ramanathan in 1923, to isolate the modified radiations, it was found that they could be readily observed in a great many liquids, and that in many cases they were strongly polarized.

Thanks to the vastly more powerful illumination made available by the 7-inch refractor, the spectroscopic examination of the effect, which had been abandoned in 1925 as indecisive, now came within the reach of direct visual study. With a Zeiss cobalt-glass filter placed in the path of the incident beam and one or other of a series of organic liquids as the scattering substance, a band in the blue-green region was observed by me in the spectrum of the scattered light, separated by a dark interval from the indigo-violet region transmitted by the filter. *Both* of these regions in the spectrum became sharper when the region of transmission was narrowed by the insertion of an additional filter in the incident beam. This suggested the employment, instead

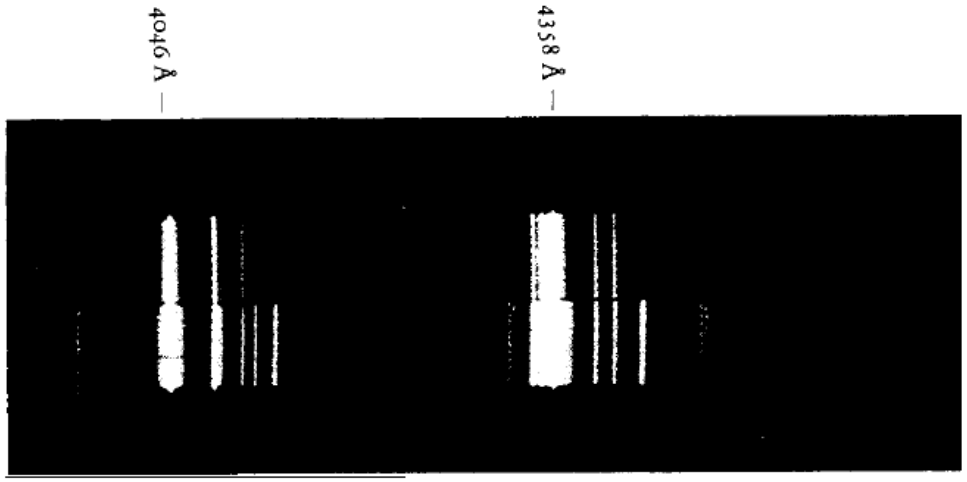


Fig. 1. Spectrum of carbon tetrachloride.

- **Raman spectroscopy** is a spectroscopic technique used to study vibrational, rotational, and other low-frequency modes in a system.

- It relies on inelastic scattering, or **Raman scattering**, of monochromatic light, usually from a **laser in the <u>visible</u>, <u>near infrared</u>, or <u>near ultraviolet</u> range**.

- The laser light interacts with molecular vibrations, phonons or other excitations in the system, resulting in the energy of the laser photons being shifted up or down.

- Raman scattering is generally weak and is 1% of the incident radiation.

A sample is illuminated with a laser beam.

Light from the illuminated spot is collected with a lens and sent through a monochromator.

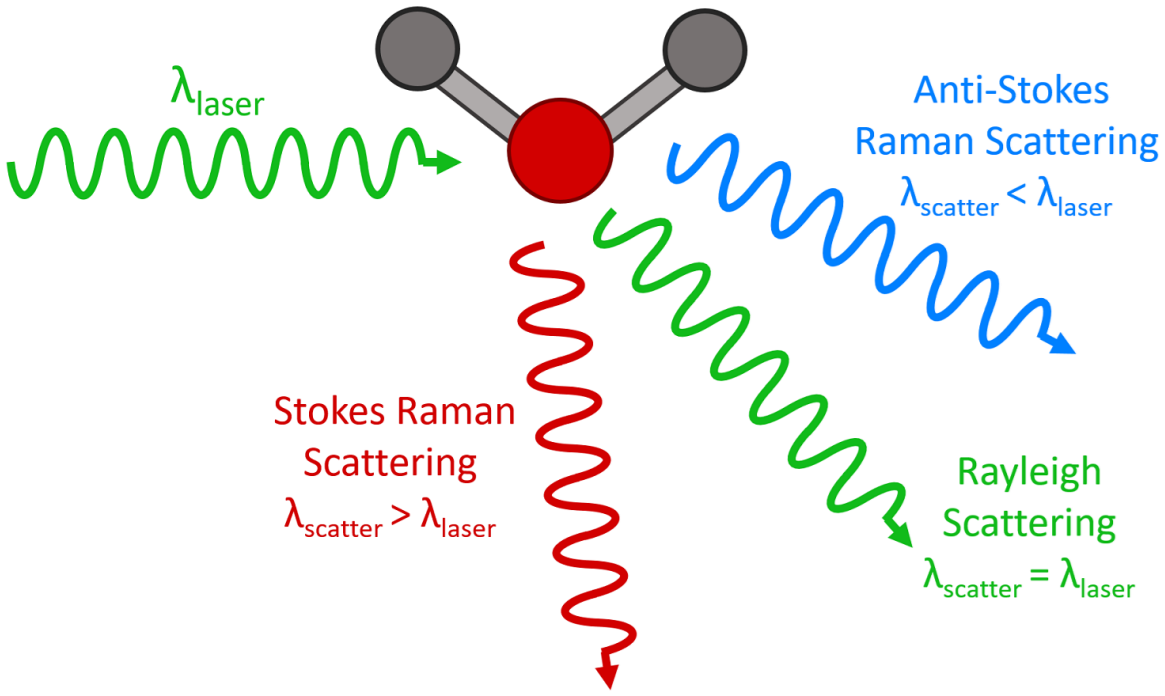
Wavelengths close to the laser line, due to elastic Rayleigh scattering, are filtered out while the rest of the collected light is dispersed onto a detector.

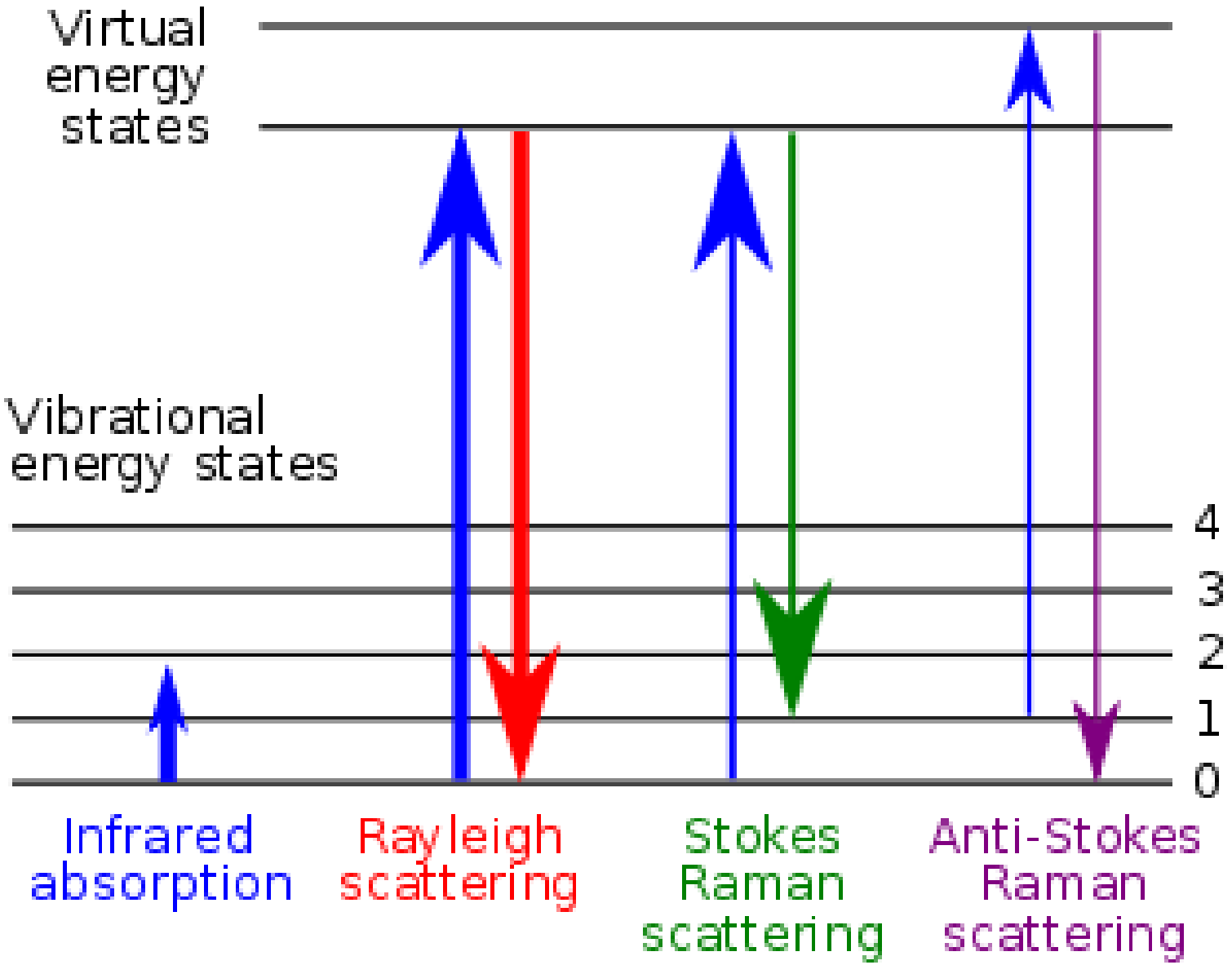
Spontaneous Raman scattering is typically very weak, and as a result the main difficulty of Raman spectroscopy is separating the weak inelastically scattered light from the intense Rayleigh scattered laser light.

- Frequency of the scattered radiation is **equal** to that of the incident radiation beam. This is known as **Rayleigh scattering**.

- Frequency of the scattered radiation is **smaller** than that of the incident radiation beam. This is known as **Stokes shift**.

- Frequency of the scattered radiation is greater than that of the incident radiation beam. This is known as **anti-Stokes shift**.

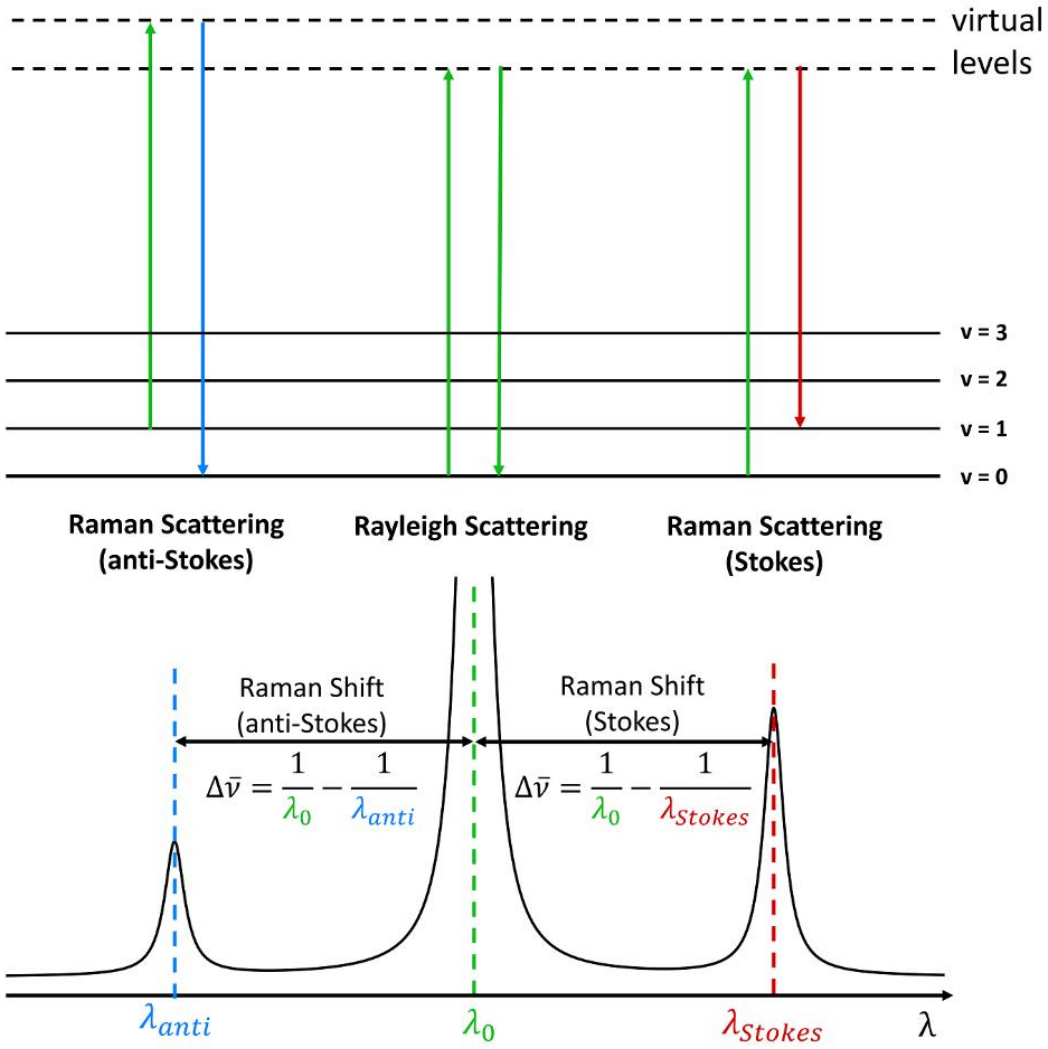



Virtual energy states
Vibrational energy states
4
3
2
1
0
Infrared absorption
Rayleigh scattering
Stokes Raman scattering
Anti-Stokes Raman scattering

- Change in the frequency of the incident radiation is due to the **inelastic collisions** between the molecules and photons.

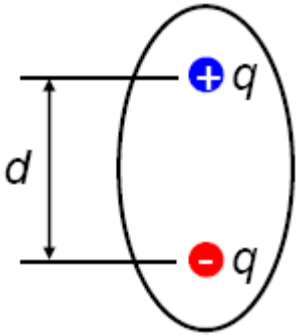
- Changes observed in the frequency of the scattered radiation depend on the nature and structure of the molecules in the sample.

- The changes observed in the frequency of radiation are called the **Raman shift**, which is the <u>difference between the frequencies of Rayleigh line and a particular Raman line</u>.



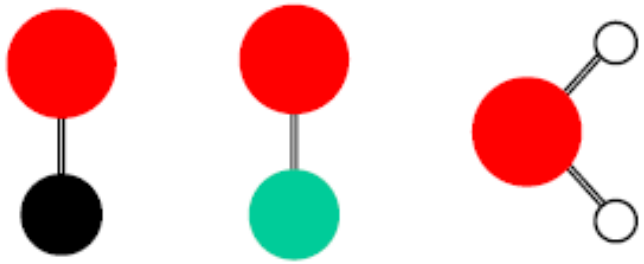
Electric dipole moment

For a single dipole with a distance, d, between charges –q and +q, the dipole moment is: $\mu = qd$



Heteronuclear diatomics and asymmetric triatomics:

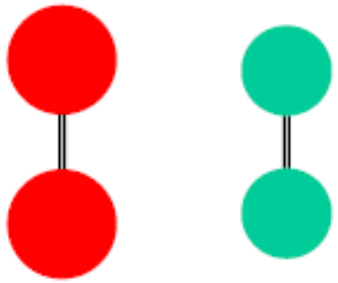
CO NO H_2O



Have permanent dipole moment

Homonuclear diatomics:

O_2 N_2

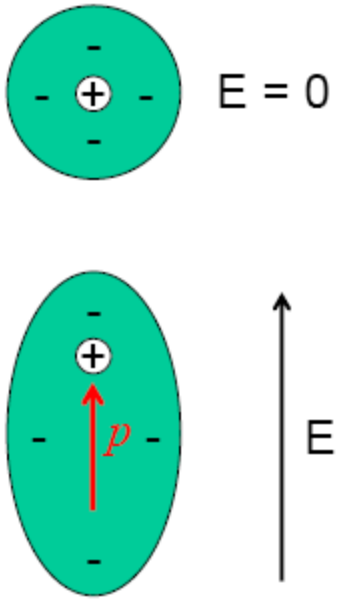


No permanent dipole moment

Induced dipole moment

A dipole moment may be created through:

- electronic excitation
- the polarizability of the molecule in the presence of an electric field (e.g. from a light wave of a laser)



Induced dipole moment:

$$\vec{\mu} = \alpha \vec{E}$$

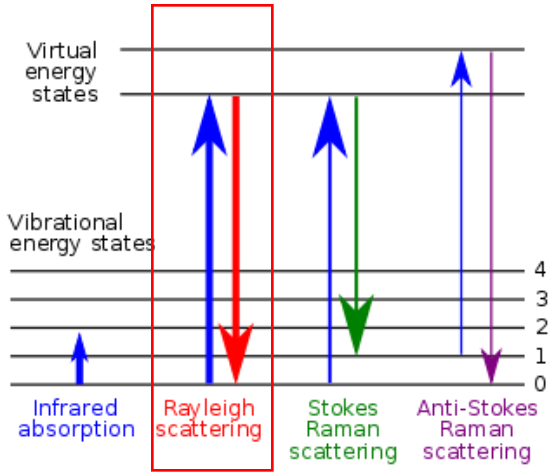
α***: molecular polarizability***

E*: electric field*

Rayleigh scattering

Rayleigh scattering is the elastic scattering of light or other electromagnetic radiation by particles much smaller than the wavelength of the light, which may be individual atoms or molecules.

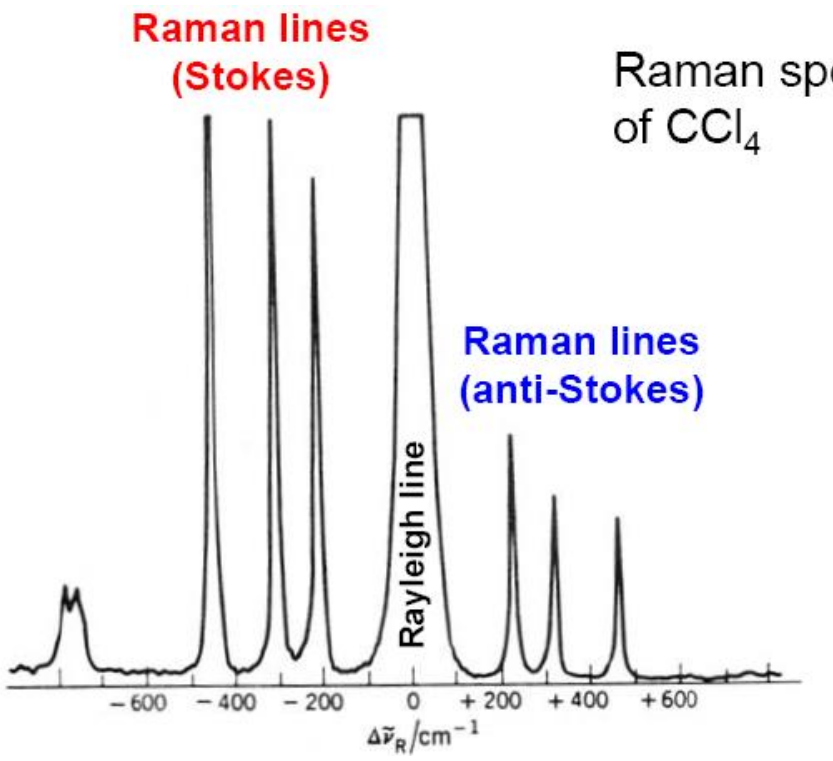
Rayleigh scattering is a function of the electric polarizability of the particles.

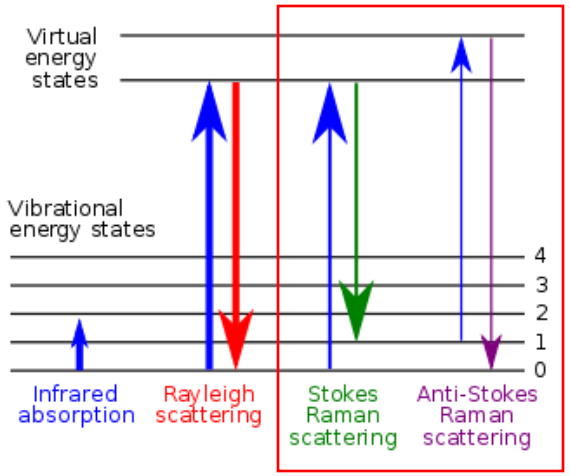


α may change as the molecule vibrates and rotates

⇒ <u>vibrational and rotational Raman scattering</u>

Raman spectrum





Pure Rotational Raman Spectra
Linear Molecules

The rotational energy levels of linear molecules:

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 \quad \text{cm}^{-1} \qquad (\text{J} = 0,1,2,....)$$

In Raman spectroscopy the polarizability of the molecule is considered.

So, the centrifugal distortion due to rotation is unimportant.

$$\varepsilon_J = BJ(J+1) \quad \text{cm}^{-1} \qquad (\text{J} = 0,1,2,....)$$

Selection Rule for Rotational Raman Spectra

The selection rule for rotational Raman spectroscopy, which states that the rotational quantum number can only change by two units ($\Delta J = \pm 2$),

- arises from the fact that a molecule's polarizability returns to its original state after rotating only 180 degrees,
    - meaning the induced distortion is effectively "seen" twice in every full rotation, leading to a dominant change of two rotational quantum numbers during scattering;
    - this is in contrast to infrared spectroscopy where the selection rule is $\Delta J = \pm 1$ due to the interaction with the changing dipole moment during rotation.

Selection Rule for Rotational Raman Spectra

$$\Delta J = 0, \text{ or } \pm 2$$

The rotational quantum number changes by two units rather than one.

$\rightarrow$ A result due to the symmetry of the polarizability ellipsoid.

Considering that the upper state quantum number is necessarily greater than the lower state quantum number,

and $\Delta J = 0$ transition does not represent any change in the molecular energy

We consider only $\Delta J = +2$ in determining the energy change due to the transition.

$$\Delta\varepsilon = \varepsilon_{J'=J+2} - \varepsilon_{J''=J} = B(4J+6) \quad \text{cm}^{-1}$$

So, $\Delta J = +2$ is labeled as *S* branch lines.

$$\Delta\varepsilon_S = B(4J+6) \quad \text{cm}^{-1} \quad (\text{J} = 0,1,2,....)$$

If a molecule **gains** rotational energy from the colliding photon $\rightarrow$ a series of **S branch** lines are generated.

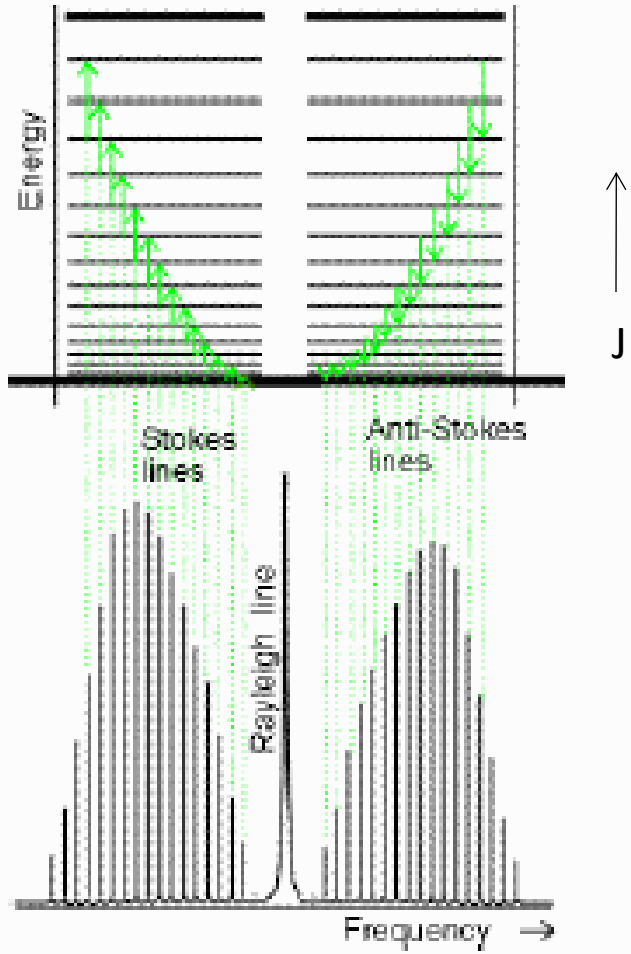
They appear to the <u>low wavenumber</u> side of the exciting line.

Stokes lines

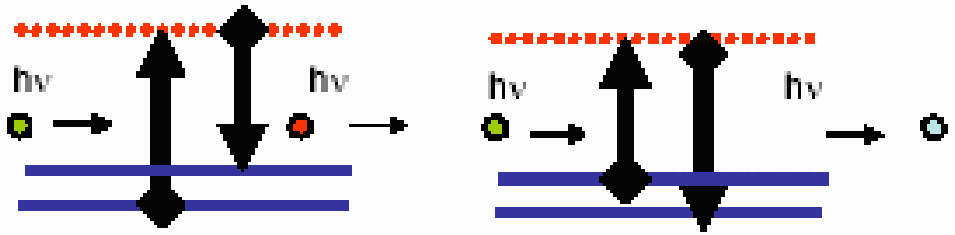
If a molecule **loses** rotational energy to the colliding photon $\rightarrow$ a series of **S branch** lines are generated.

They appear to the <u>higher wavenumber</u> side of the exciting line.

anti-Stokes lines



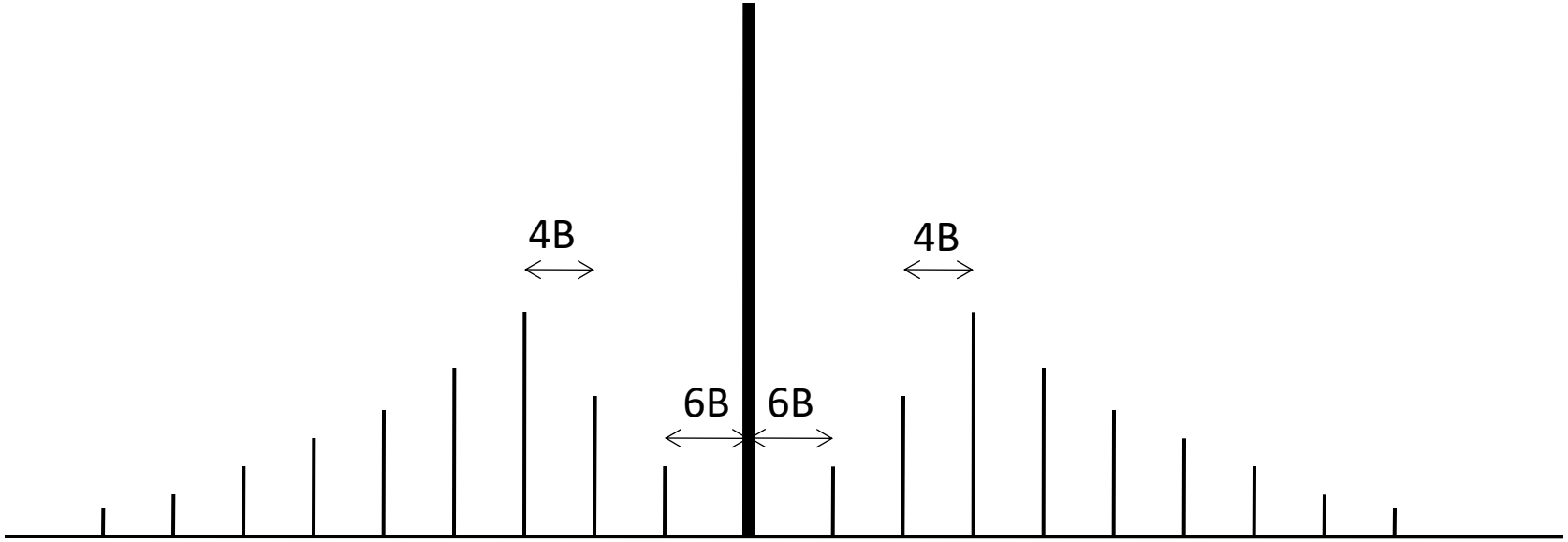
Stokes **Anti-Stokes**



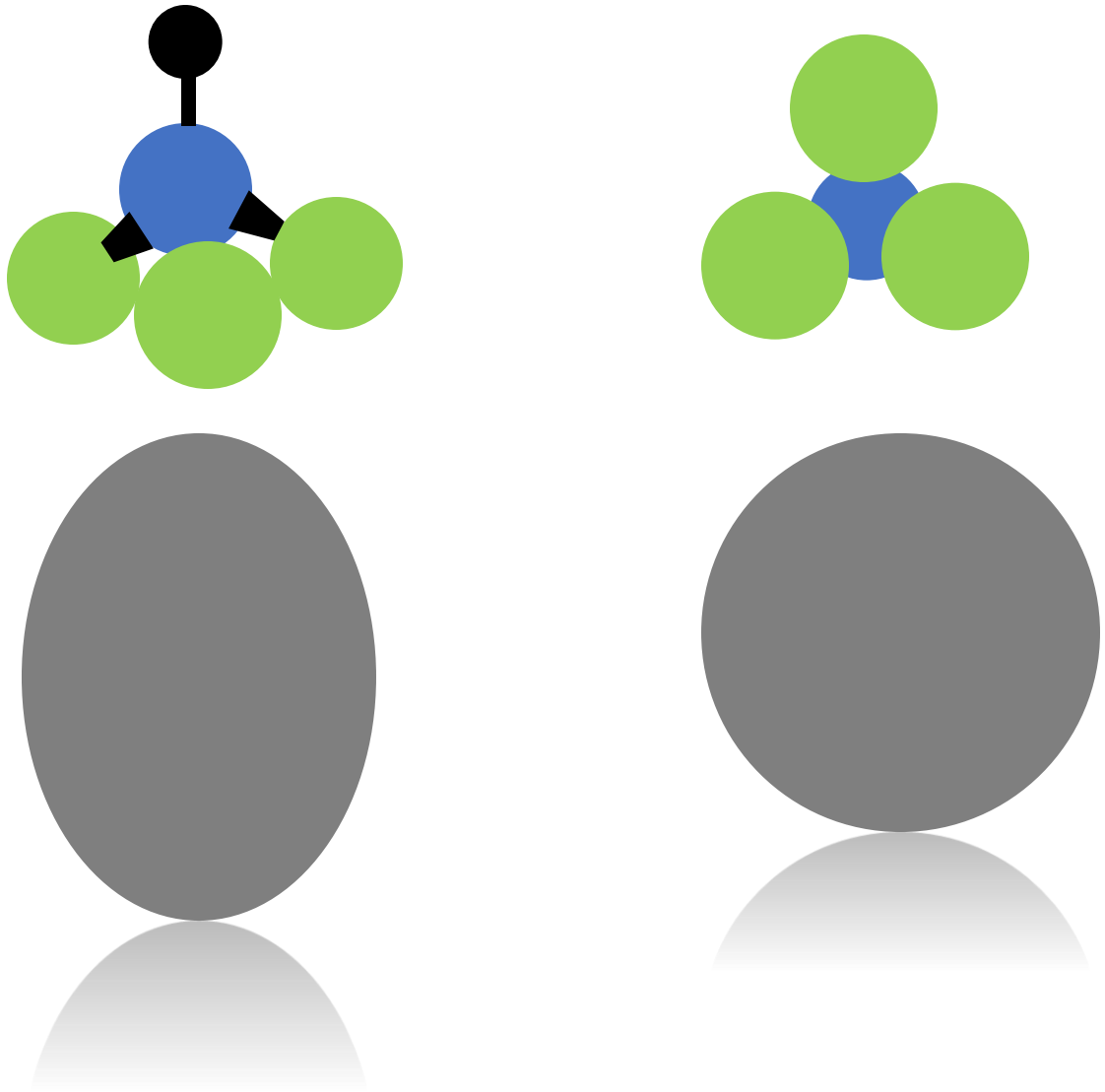
$$\Delta\varepsilon_S = B(4J+6) \quad \text{cm}^{-1} \quad (\text{J} = 0,1,2,....)$$

<u>Putting J = 0</u> $\rightarrow$ Separation of the first line from the exciting line is **6B cm^{-1}**

$\rightarrow$ Separation of the successive lines is **4B cm^{-1}**



Symmetric Top Molecules



Chloroform molecule and its polarizability ellipsoid from across and along the symmetry axis.

Rotation about the top axis does not produce any change in the polarizability.

End-over-end rotation produces a change in the polarizability.

The energy levels of a symmetric top molecule is given by:

$$\varepsilon_{J,K} = BJ(J+1) + (A-B)K^2 \quad \text{cm}^{-1} \quad (\text{J} = 0,1,2,\ldots;\ \text{K} = \pm\text{J}, \pm(\text{J}-1),\ldots)$$

The Selection Rules for Raman spectra are:

ΔK = 0 and ΔJ = 0, $\pm$1, $\pm$2

K is the rotational quantum number for axial rotation.

ΔK = 0 implies that changes in the angular momentum about the top axis will not give rise to a Raman spectrum.

For positive ΔJ, we have:

Case 1: ΔJ = +1 (R branch lines)

$$\Delta\varepsilon_R = \varepsilon_{J'=J+1} - \varepsilon_{J''=J} = 2B(J+1) \quad \text{cm}^{-1} \qquad (\text{J} = 1,2,3,...; \text{J} \neq 0)$$

Case 2: ΔJ = +2 (S branch lines)

$$\Delta\varepsilon_S = \varepsilon_{J'=J+2} - \varepsilon_{J''=J} = B(4J+6) \quad \text{cm}^{-1} \qquad (\text{J} = 0,1,2,3,...)$$

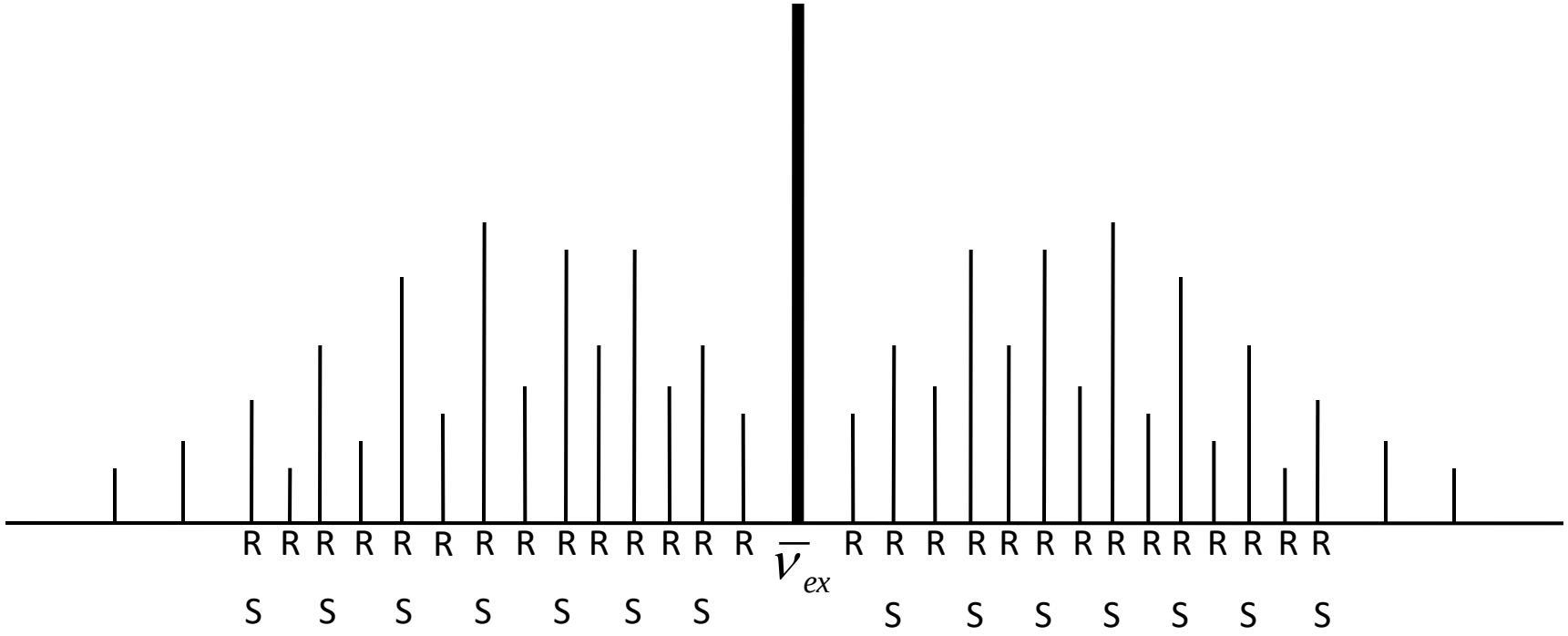
So, two series of lines in the Raman spectrum.

$$\bar{\nu}_R = \bar{\nu}_{ex} \pm \Delta\varepsilon_R = \bar{\nu}_{ex} \pm 2B\left(J+1\right) \quad \text{cm}^{-1} \qquad (\text{J} = 1,2,....)$$

$$\bar{\nu}_S = \bar{\nu}_{ex} \pm \Delta\varepsilon_S = \bar{\nu}_{ex} \pm B\left(4J+6\right) \quad \text{cm}^{-1} \qquad (\text{J} = 0,1,2,....)$$

$$\bar{\nu}_R = \bar{\nu}_{ex} \pm \Delta\varepsilon_R = \bar{\nu}_{ex} \pm 2B(J+1) \quad \text{cm}^{-1} \qquad (\text{J} = 1,2,....)$$

$$\bar{\nu}_S = \bar{\nu}_{ex} \pm \Delta\varepsilon_S = \bar{\nu}_{ex} \pm B(4J+6) \quad \text{cm}^{-1} \qquad (\text{J} = 0,1,2,....)$$



The Rotational Raman Spectrum of a Symmetric Top Molecule